Project: Worked examples
Title: Press and shrink fit
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1. Scope and method

A shaft (solid or hollow) pressed or shrunk into a hub: the interface pressure for a given diametral interference, the hoop and radial stresses at the interface in hub and shaft, von Mises utilisation against each yield strength, the axial assembly force and the torque the joint can carry by friction, and the temperature rise to shrink-fit the hub.

WHAT IT CALCULATES	RELATION
Radial interference	$\delta_r = \delta/2$
Interface pressure, the hub and the shaft as thick cylinders	$p = \delta_r / [\ (R/E_h)\cdot((r_o^2 + R^2)/(r_o^2 - R^2) + \nu_h) + (R/E_s)\cdot((R^2 + r_i^2)/(R^2 - r_i^2) - \nu_s)\ ]$
Stresses at the hub bore	$\sigma_\theta = p\cdot(r_o^2 + R^2)/(r_o^2 - R^2)$; $\sigma_r = -p$
Stresses at the shaft surface	$\sigma_\theta = -p\cdot(R^2 + r_i^2)/(R^2 - r_i^2)$; $\sigma_r = -p$
Surface smoothing on assembly (when Rz is entered)	$\delta_{eff} = \max(\delta - 0.8\cdot(Rz_h + Rz_s), 0)$, $p_{eff} = p\cdot\delta_{eff}/\delta$
Axial force and torque the fit holds by friction, on p_{eff}	$F = \mu\cdot p_{eff}\cdot 2\pi R\cdot L$; $T = \mu\cdot p_{eff}\cdot 2\pi R^2\cdot L$. The stresses stay at the nominal p (the larger interference of the fit)
Temperature difference to assemble by heating the hub, with assembly clearance c_a	$\Delta T = (\delta + c_a)/(\alpha\cdot 2R)$

Basis: Press and shrink fits; stresses in pressurised cylinders[1](§3-16, §3-17); Thick-walled cylinders under pressure[2](Ch. 13, Table 13.5).

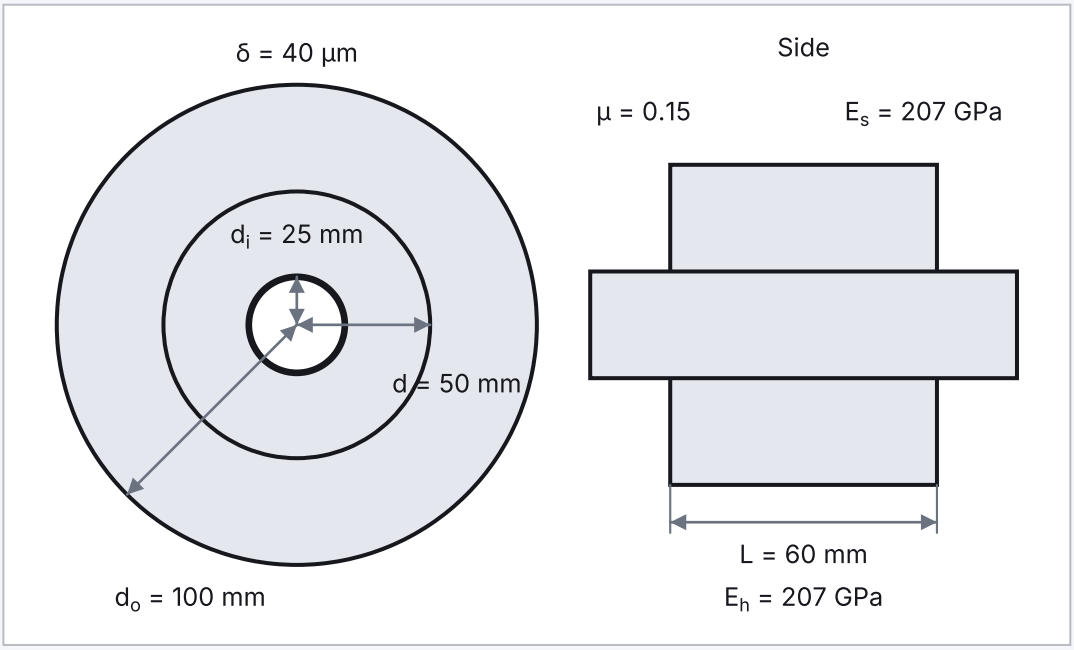


Figure 1 — A hub shrunk onto a shaft: section across the interface, and the engagement length beside it. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Interface (nominal) diameter	50	mm	drawing: DRG-HUB-04 rev C	1 … 1000 mm
δ	Diametral interference	40	µm	drawing: DRG-HUB-04 rev C	1 … 500 µm
d_o	Hub outer diameter	100	mm	drawing: DRG-HUB-04 rev C	1 … 2000 mm
d_i	Shaft bore diameter (0 for a solid shaft)	25	mm	drawing: DRG-SHAFT-13 rev A	0 … 1000 mm
L	Engagement length	60	mm	drawing: DRG-HUB-04 rev C	1 … 2000 mm
E_h	Hub Young's modulus	207	GPa	spec: steel	10 … 450 GPa
ν_h	Hub Poisson's ratio	0.292		spec: steel	0.1 … 0.45
E_s	Shaft Young's modulus	207	GPa	spec: steel	10 … 450 GPa
ν_s	Shaft Poisson's ratio	0.292		spec: steel	0.1 … 0.45
µ	Coefficient of friction at the interface	0.15		assumption: Dry steel on steel.	0.05 … 0.4
$S_{y,h}$	Hub yield strength	355	MPa	spec: EN 10025 S355	50 … 2000 MPa
$S_{y,s}$	Shaft yield strength	355	MPa	spec: EN 10025 S355	50 … 2000 MPa
α_h	Hub thermal expansion coefficient	0.000012	1/K	spec: steel	0.000005 … 0.00003 1/K
c_a	Assembly clearance wanted for shrink fitting	20	µm	assumption: Assembly clearance.	0 … 500 µm

3. Assumptions

A1. Both parts are long cylinders in plane stress; the interface pressure is uniform along the engagement. (Lamé thick-cylinder relations; end effects neglected.)
A2. The interference is the nominal one given: the torque and the press force need the minimum interference of the fit, the stresses the maximum, and the surface roughness smooths on assembly and takes about $0.8 \cdot (Rz_h + Rz_s)$ off it unless the roughness is entered. (DIN 7190-1; enter Rz to apply the smoothing to the torque and the force.)
A3. The friction coefficient is as entered and the same for assembly and for torque transmission. (Enter it with its source; it dominates the force and torque results.)
A4. Yield is checked by the von Mises equivalent stress at the hub bore, at the shaft surface and, for a hollow shaft, at the shaft bore (where the hoop stress is largest: $2 \cdot p \cdot R^2/(R^2 - r_i^2)$, with no radial stress), the larger of the two shaft values governing. (Axial stress taken as zero.)
A5. µ (Coefficient of friction at the interface) is taken as 0.15. (Dry steel on steel.)
A6. c_a (Assembly clearance wanted for shrink fitting) is taken as 20 µm. (Assembly clearance.)
A7. No surface smoothing is applied: the torque and the press force are at the nominal interference. Roughness takes about $0.8 \cdot (Rz_h + Rz_s)$ off it on assembly, so a fine fit grips less than shown; check the torque at the minimum interference of the fit and the stresses at the maximum. (Enter Rz_h and Rz_s to apply the smoothing (DIN 7190-1).)

4. Working

[15] $R = d / 2$
$= 50\ \text{mm} / 2$
$= 25\ \text{mm}$
Interface radius

[16] $r_o = d_o / 2$
$= 100\ \text{mm} / 2$
$= 50\ \text{mm}$
Hub outer radius

[17] $r_i = d_i / 2$
$= 25\ \text{mm} / 2$
$= 12.5\ \text{mm}$
Shaft bore radius

[18] $\delta_r = \delta / 2$
$= 40\ \mu\text{m} / 2$
$= 20\ \mu\text{m}$
Radial interference

[19] $C_h = (R / E_h) * ((r_o^2 + R^2) / (r_o^2 - R^2) + \nu_h)$ [1](§3-16)
$= 25\ \text{mm} / 207\ \text{GPa} \times (((50\ \text{mm})^2 + (25\ \text{mm})^2) / ((50\ \text{mm})^2 - (25\ \text{mm})^2) + 0.292)$
$= 0.23655\ \text{mm/GPa}$
Hub compliance

[20] $C_s = (R / E_s) * ((R^2 + r_i^2) / (R^2 - r_i^2) - \nu_s)$ [1](§3-16)
$= 25\ \text{mm} / 207\ \text{GPa} \times (((25\ \text{mm})^2 + (12.5\ \text{mm})^2) / ((25\ \text{mm})^2 - (12.5\ \text{mm})^2) - 0.292)$
$= 0.16602\ \text{mm/GPa}$
Shaft compliance

[21] $p = \delta_r / (C_h + C_s)$ [1](eq. 3-56)
$= 20\ \mu\text{m} / (0.23655\ \text{mm/GPa} + 0.16602\ \text{mm/GPa})$
$= 49.68\ \text{MPa}$
Interface pressure

[22] $\sigma_{t,hub} = p * (r_o^2 + R^2) / (r_o^2 - R^2)$ [1](eq. 3-57)
$= 49.68\ \text{MPa} \times ((50\ \text{mm})^2 + (25\ \text{mm})^2) / ((50\ \text{mm})^2 - (25\ \text{mm})^2)$
$= 82.8\ \text{MPa}$
Hub hoop stress at the bore

[23] $\sigma_{r,hub} = -p$
$= -49.68\ \text{MPa}$
$= -49.68\ \text{MPa}$
Hub radial stress at the bore

[24] $\sigma_{t,shaft} = -p * (R^2 + r_i^2) / (R^2 - r_i^2)$ [1](eq. 3-58)
$= -49.68\ \text{MPa} \times ((25\ \text{mm})^2 + (12.5\ \text{mm})^2) / ((25\ \text{mm})^2 - (12.5\ \text{mm})^2)$
$= -82.8\ \text{MPa}$
Shaft hoop stress at the surface

[25] $\sigma_{r,shaft} = -p$
$= -49.68\ \text{MPa}$
$= -49.68\ \text{MPa}$
Shaft radial stress at the surface

[26] $\sigma_{vm,hub} = \text{sqrt}(\sigma_{t,hub}^2 - \sigma_{t,hub} * \sigma_{r,hub} + \sigma_{r,hub}^2)$
$= \text{sqrt}((82.8\ \text{MPa})^2 - 82.8\ \text{MPa} \times (-49.68\ \text{MPa}) + (-49.68\ \text{MPa})^2)$
$= 115.92\ \text{MPa}$
Hub von Mises stress at the bore

[27] $\sigma_{vm,shaft} = \text{sqrt}(\sigma_{t,shaft}^2 - \sigma_{t,shaft} * \sigma_{r,shaft} + \sigma_{r,shaft}^2)$
$= \text{sqrt}((-82.8\ \text{MPa})^2 - (-82.8\ \text{MPa}) \times (-49.68\ \text{MPa}) + (-49.68\ \text{MPa})^2)$
$= 72.183\ \text{MPa}$
Shaft von Mises stress at the surface

[28] $\sigma_{t,shaft,bore}$ = -2 * p * R^2 / (R^2 - r_i^2) [3](cylinder)
= -2 × 49.68 MPa × (25 mm)^2 / ((25 mm)^2 - (12.5 mm)^2)
= -132.48 MPa
Shaft hoop stress at the shaft bore (hollow shaft; the radial stress is zero there)
[29] $\sigma_{vm,shaft,bore}$ = abs($\sigma_{t,shaft,bore}$)
= abs(-132.48 MPa)
= 132.48 MPa
Shaft von Mises stress at the shaft bore (hollow shaft: hoop stress alone)
[30] $\sigma_{vm,shaft,max}$ = max($\sigma_{vm,shaft}$, $\sigma_{vm,shaft,bore}$)
= max(72.183 MPa, 132.48 MPa)
= 132.48 MPa
Largest shaft von Mises stress, surface or bore
[31] δ_{eff} = δ
= 40 µm
= 40 µm
Effective interference (no roughness entered: the nominal interference)
[32] p_{eff} = p * δ_{eff} / δ
= 49.68 MPa × 40 µm / 40 µm
= 49.68 MPa
Interface pressure that grips, at the effective interference
[33] F_a = µ * p_{eff} * 2 * pi * R * L
= 0.15 × 49.68 MPa × 2 × π × 25 mm × 60 mm
= 70.233 kN
Axial assembly (press) force
[34] T = µ * p_{eff} * 2 * pi * R^2 * L
= 0.15 × 49.68 MPa × 2 × π × (25 mm)^2 × 60 mm
= 1755.8 N·m
Torque capacity by friction
[35] dT = (δ + c_a) / (α_h * d)
= (40 µm + 20 µm) / (0.000012 1/K × 50 mm)
= 100 K
Hub temperature rise for shrink fitting

5. Results

SYMBOL	VALUE	DESCRIPTION
p	49.68 MPa	Interface pressure
$\sigma_{t,hub}$	82.8 MPa	Hub hoop stress at bore
$\sigma_{t,shaft}$	-82.8 MPa	Shaft hoop stress at surface
$\sigma_{vm,hub}$	115.92 MPa	Hub von Mises stress
$\sigma_{vm,shaft}$	72.183 MPa	Shaft von Mises stress at the surface
$\sigma_{vm,shaft,bore}$	132.48 MPa	Shaft von Mises stress at the shaft bore
$\sigma_{vm,shaft,max}$	132.48 MPa	Largest shaft von Mises stress (surface or bore)
δ_{eff}	40 µm	Effective interference after smoothing
p_{eff}	49.68 MPa	Gripping interface pressure
F_a	70.233 kN	Assembly force
T	1755.8 N·m	Torque capacity
dT	100 K	Shrink-fit temperature rise

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Shaft yield, surface and bore	132.48 MPa	355 MPa	0.373	
PASS	Hub yield	115.92 MPa	355 MPa	0.327	

7. Applicability

No warnings.

8. Verification

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Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
3. Lamé, external pressure on a hollow.